

Algorithms for Wireless Communications

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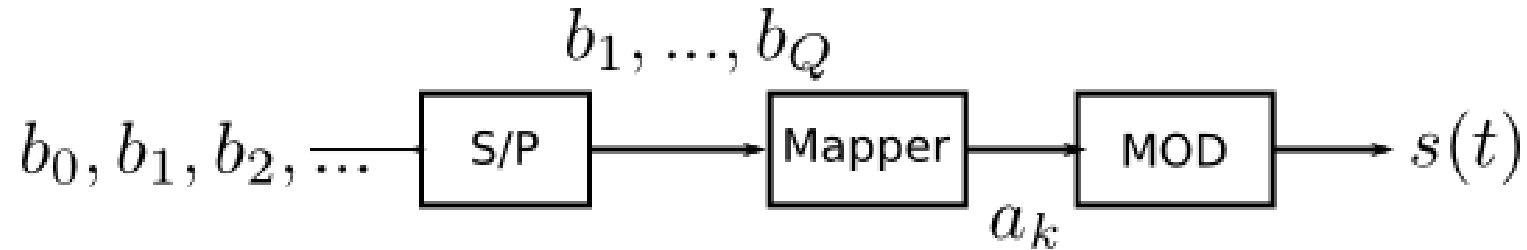
Passband Modulation

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Modulation with Continuous Time Signals

The discrete time representation is an abstract model of a communication system for analysis and for signal processing

- Communication over real-world channels happens using *continuous time signals*



- Data is mapped to subsequent symbols

T : Symbol (baud) rate

- MOD:** Different symbols are represented by (different) cont. time waveforms

modulation methods	linear	nonlinear
memoryless	PAM (QAM MPSK)	
with memory		continuous phase FSK, MSK, CPM

Linear Modulation

“Mathematical relationship between data symbols and the modulated signal is a linear function”

Signal model

$$s(t) = \sum_{k=-\infty}^{\infty} s_{a_k}(t - kT)$$

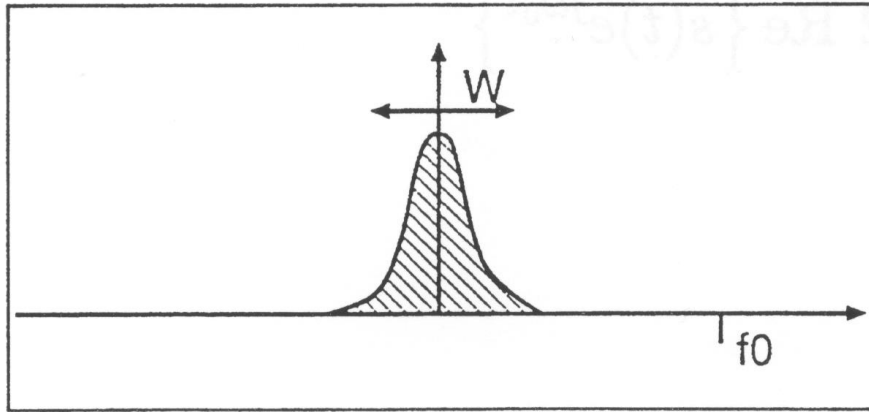
Lowpass vs. Bandpass Transmission

Consider the Spectrum of a signal

$$s(t) \xrightarrow{\mathcal{F}\{\}} S(f)$$

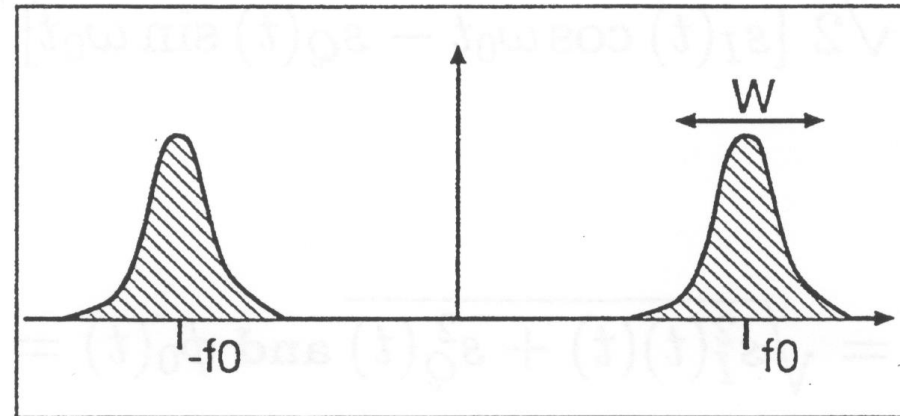
Low-pass transmission

Spectrum centered around DC



Bandpass transmission

Spectrum centered around a carrier frequency f_0



Physical signals are always real-valued: Spectrum is complex-conjugate symmetric

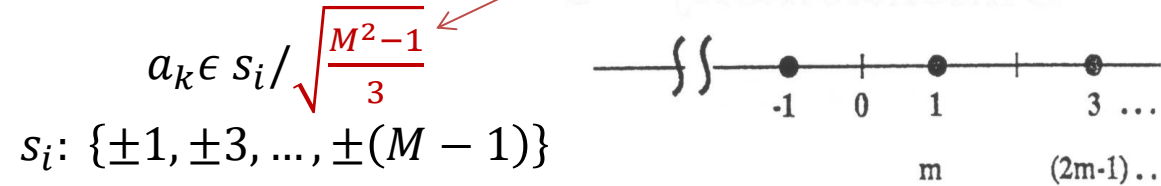
$$S(-f) = S^*(f)$$

Linear Baseband Modulation for Baseband Communication

- Based on real-valued low-pass signals

Pulse Amplitude Modulation ($M - PAM$)

- Real-valued symbol alphabet

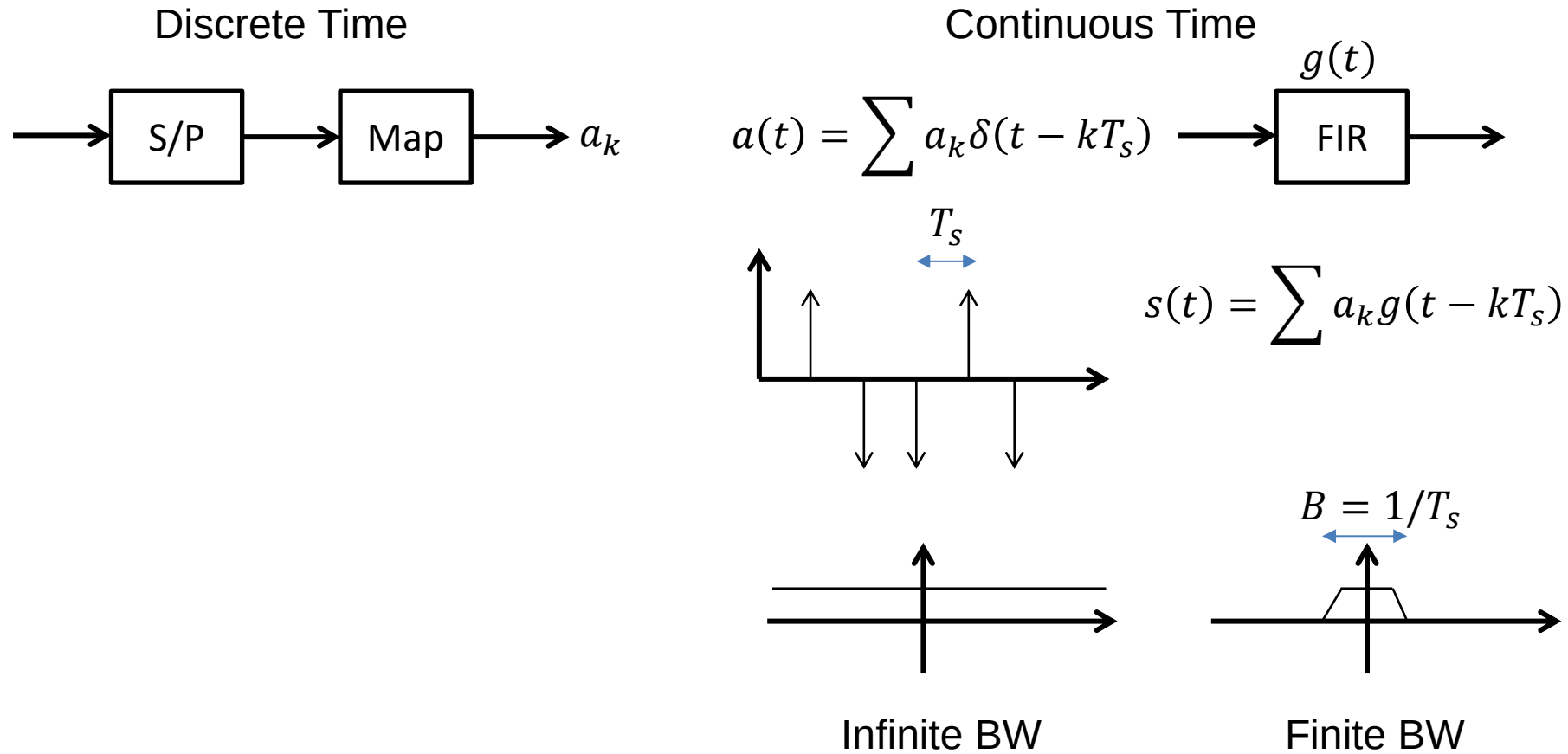


- Prototype waveform (pulse shape) $g(t)$ scaled by the data symbol a_k

PAM transmitted signal

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT)$$

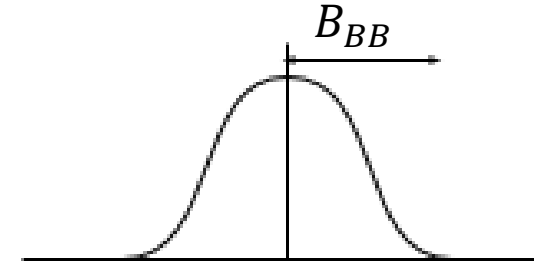
Linear BB Modulation



Baseband Communication

- Required minimum baseband bandwidth for a symbol (baud) rate T

$$B_{BB} \geq \frac{1}{2T} \text{ [Hz]}$$

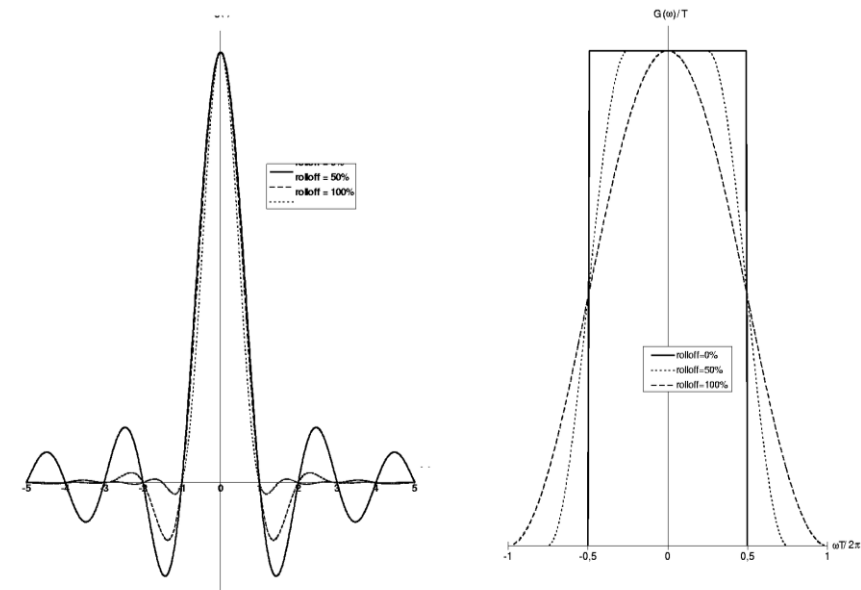


Actual bandwidth depends on the pulse shape $g(t)$

- Optimum in terms of bandwidth efficiency

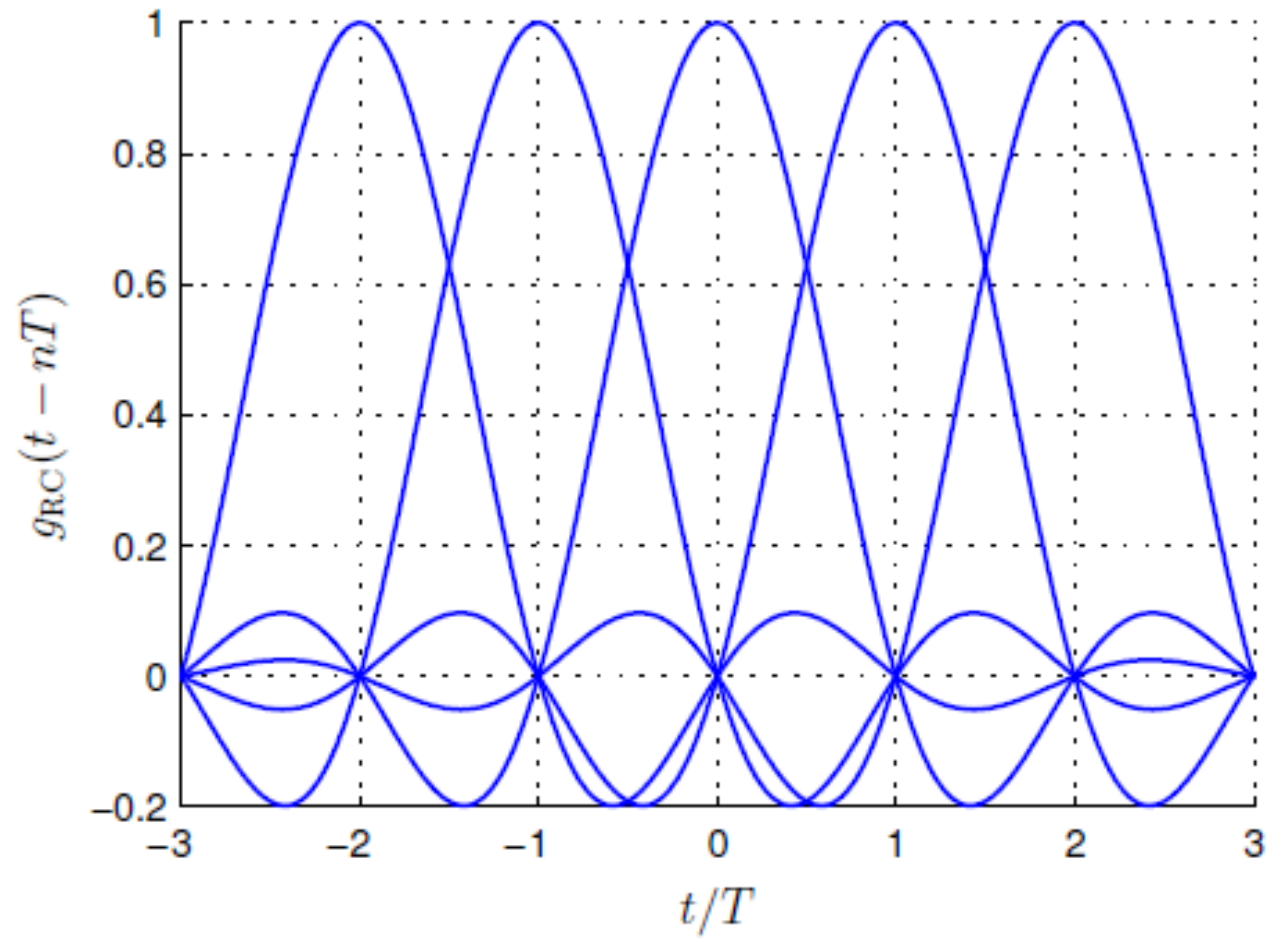
$$\text{SINC Pulse : } g(t) = \frac{\sin(\pi t/T)}{\pi t/T}$$

- Other pulse shapes (e.g., Raised Cosine) provide tradeoff between excess bandwidth and time domain behavior



Source: CDMA, Principles of Spread Spectrum Communication, Andrew J. Viterbi

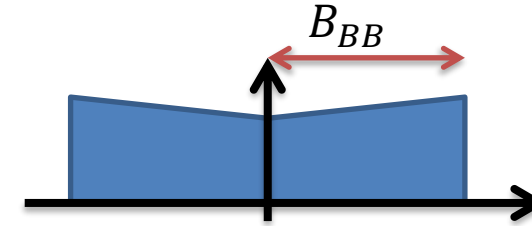
Nyquist Sampling



Baseband Signals and Transmission

Baseband signals are

- Centered around DC
- Have a lowpass characteristic

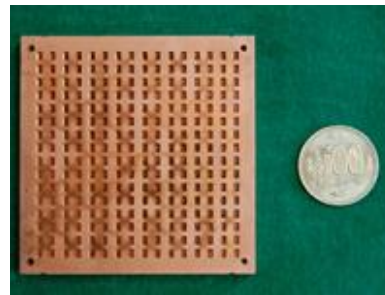


Baseband transmission is not an option for wireless systems

- Complex-valued baseband signals: no direct physical representation
- Compact antenna design difficult for low frequencies
 - Good radiation efficiency: dimension proportional to signal frequency
 - Low frequencies large antenna structures
 - High frequencies: small structures
 - Baseband signals: almost impossible to achieve matching over entire bandwidth



350 kHz antenna



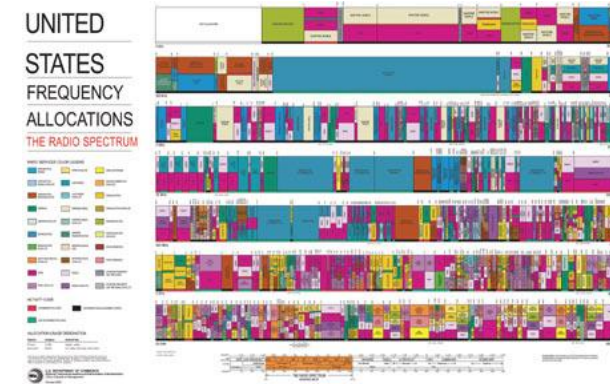
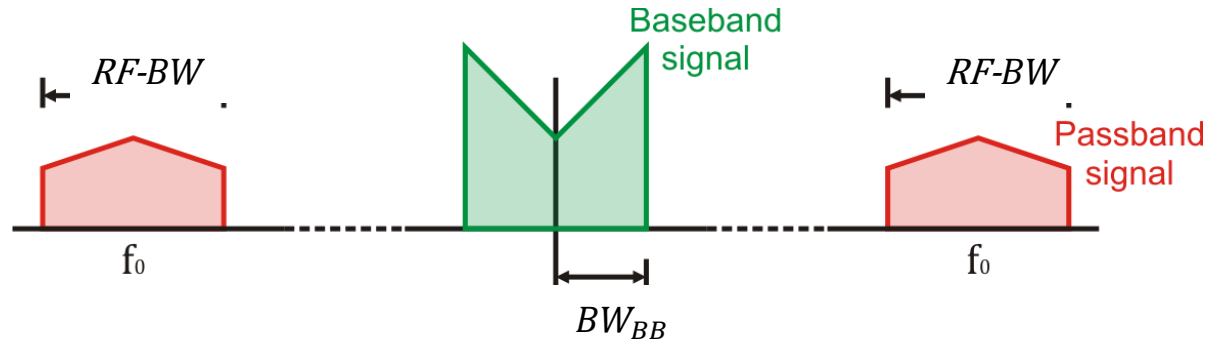
16x16 60 GHz antennas

$$\lambda \sim \frac{c}{f_c - \frac{BW}{2}} \leftrightarrow \lambda \sim \frac{c}{f_c + \frac{BW}{2}}$$

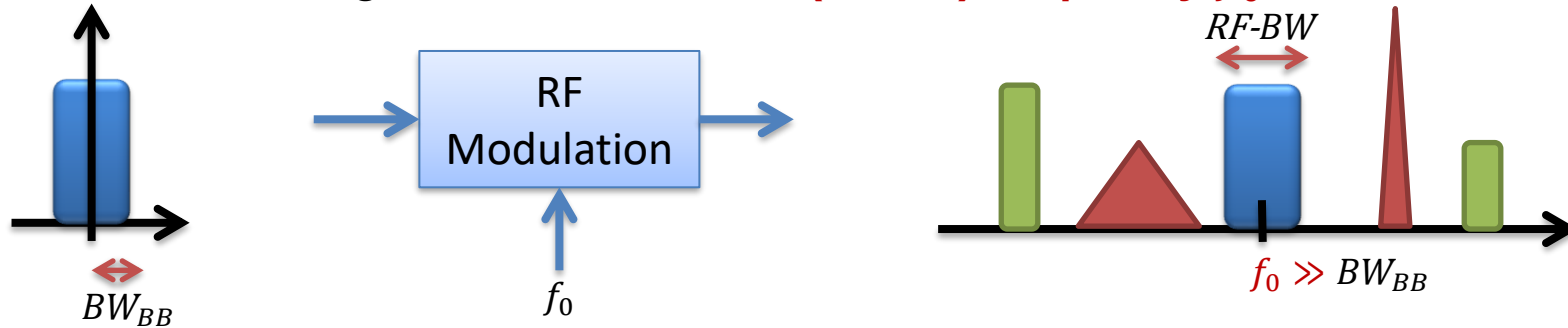
$$d \sim \lambda/4 = c/4f$$

Passband Modulation

- Spectrum is a shared resource
 - Need to separate users by using different frequency slots



Conversion of BB signal to an RF **carrier (center) frequency** f_0



- Wavelength remains almost constant over entire frequency band
 - Enables compact and efficient antennas over entire bandwidth

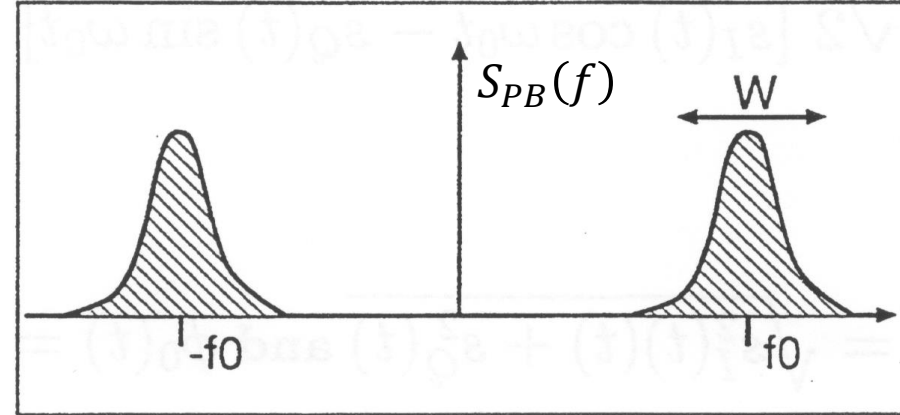
$$\lambda \sim \frac{1}{f_0 \pm \frac{BW}{2}} \approx \frac{1}{f_0} \text{ for } f_0 \gg BW_{BB}$$

Complex Baseband Representation of Passband Signals

Passband (=bandpass) signal with frequency domain representation $S_{PB}(f)$

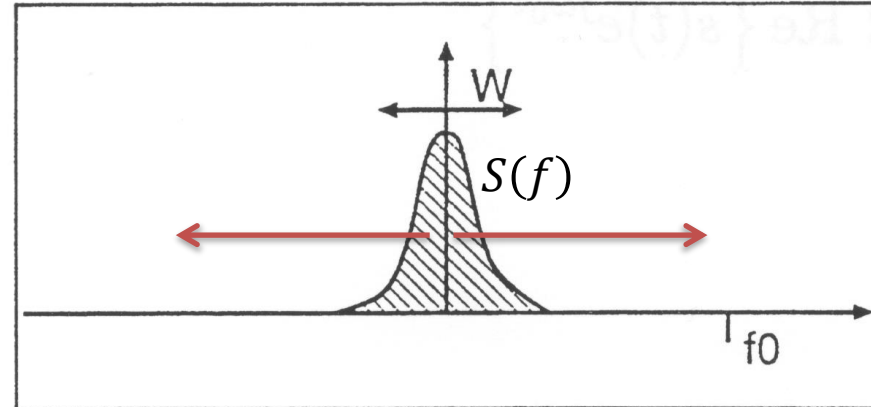
- Carrier frequency f_0
- Bandwidth is limited to W
- Bandpass signal is real-valued

$$\Rightarrow S_{PB}(-f) = S_{PB}^*(f) !!$$



Introduce the (band limited) **baseband signal** $S(f)$ so that

$$S_{BP}(f) = \frac{1}{\sqrt{2}} [S(f - f_0) + S^*(-f - f_0)]$$



Since $S(f)$ contains all the information of the bandpass signal,
we can work with $S(f)$ instead of $S_{PB}(f)$

Complex Baseband Representation

NOTE:

- $S_{PB}(f)$ is usually not symmetric around f_0
- Hence $S(-f) \neq S^*(f)$

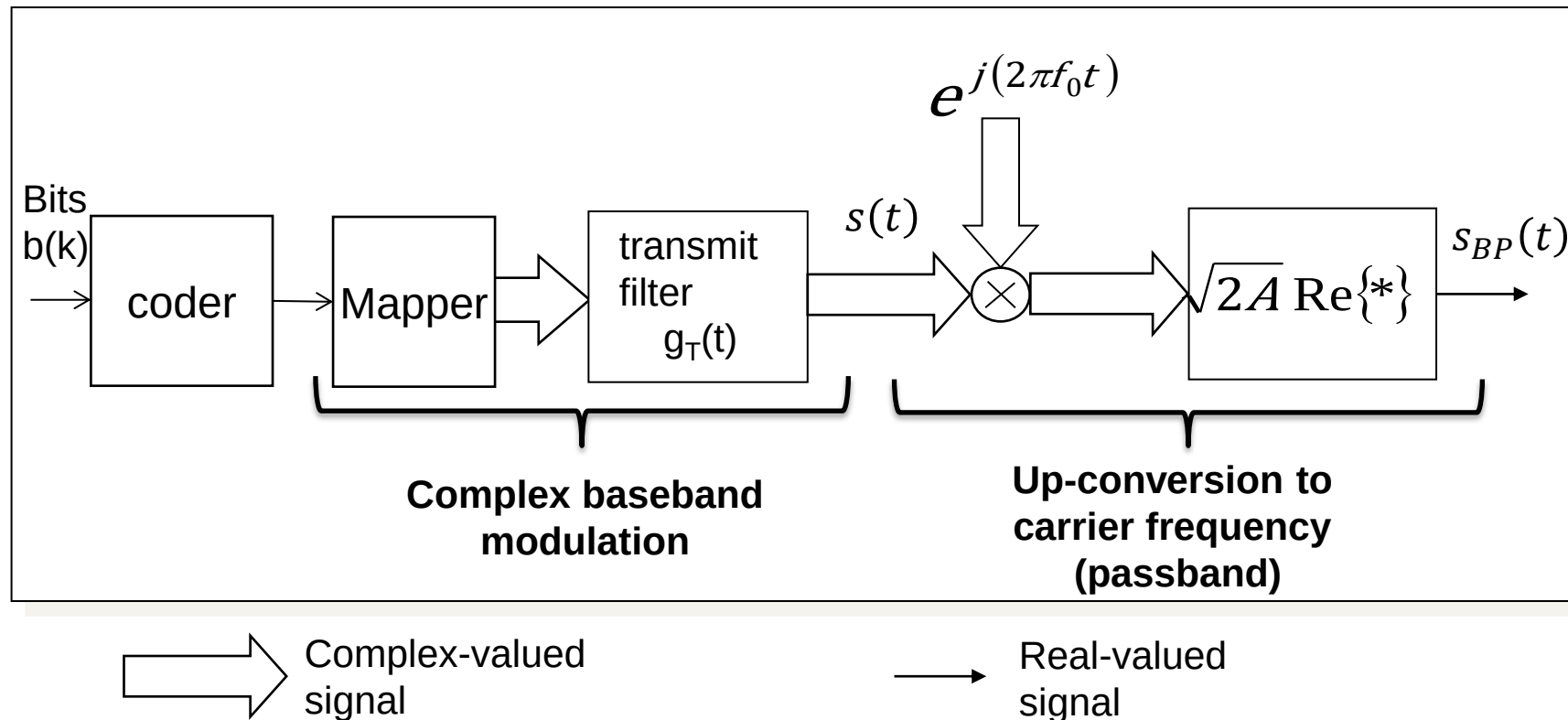
Complex-valued time domain representation of the baseband equivalent of a bandpass signal

$$s(t) = s_R(t) + js_I(t)$$

Linear Passband Modulation

Passband modulation is typically realized as a 2-step process

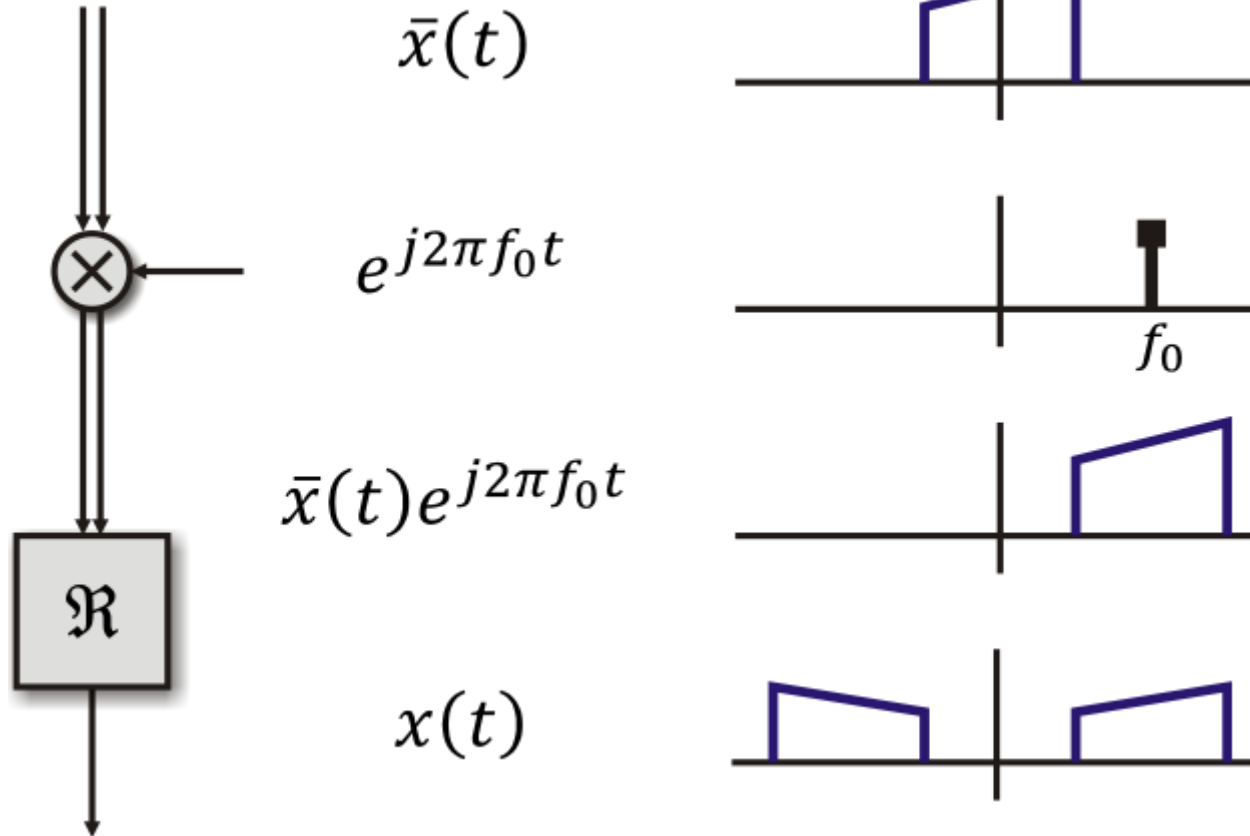
1. Baseband signal generation with linear modulation (similar to baseband communication, but complex-valued)
2. Up-conversion to the carrier frequency (passband)



Direct Conversion of Analog Bandlimited Signals

$$x(t) = \Re\{\bar{x}(t)e^{j2\pi f_0 t}\}$$

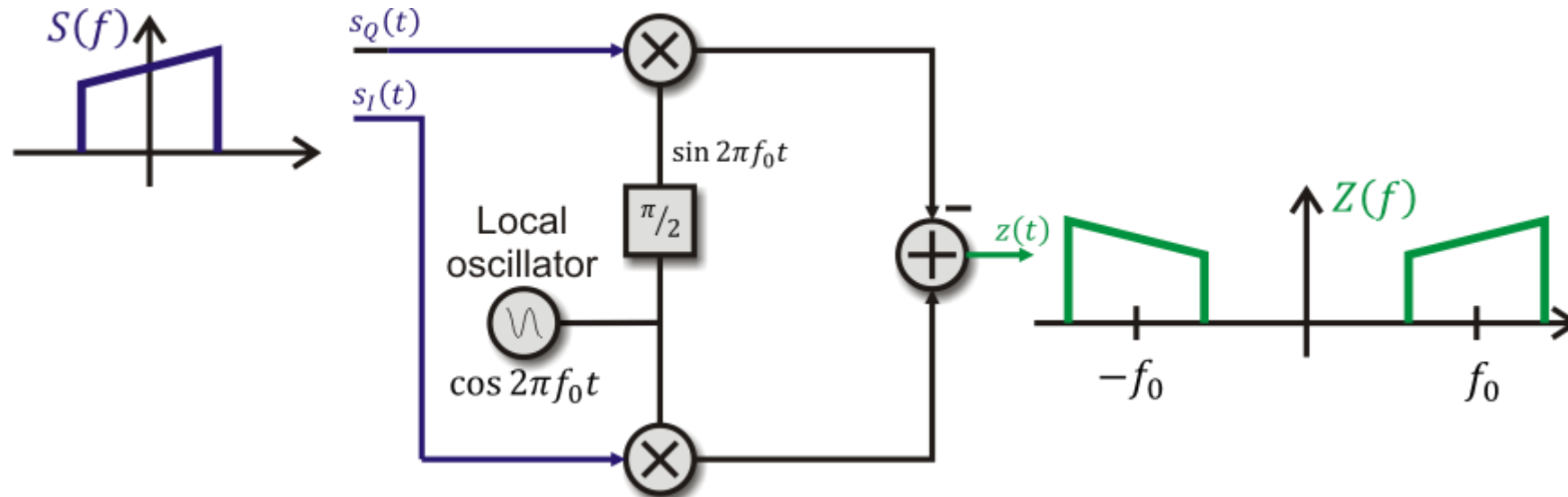
- Complex baseband signal
- RF carrier
- Modulated signal



Direct conversion architecture

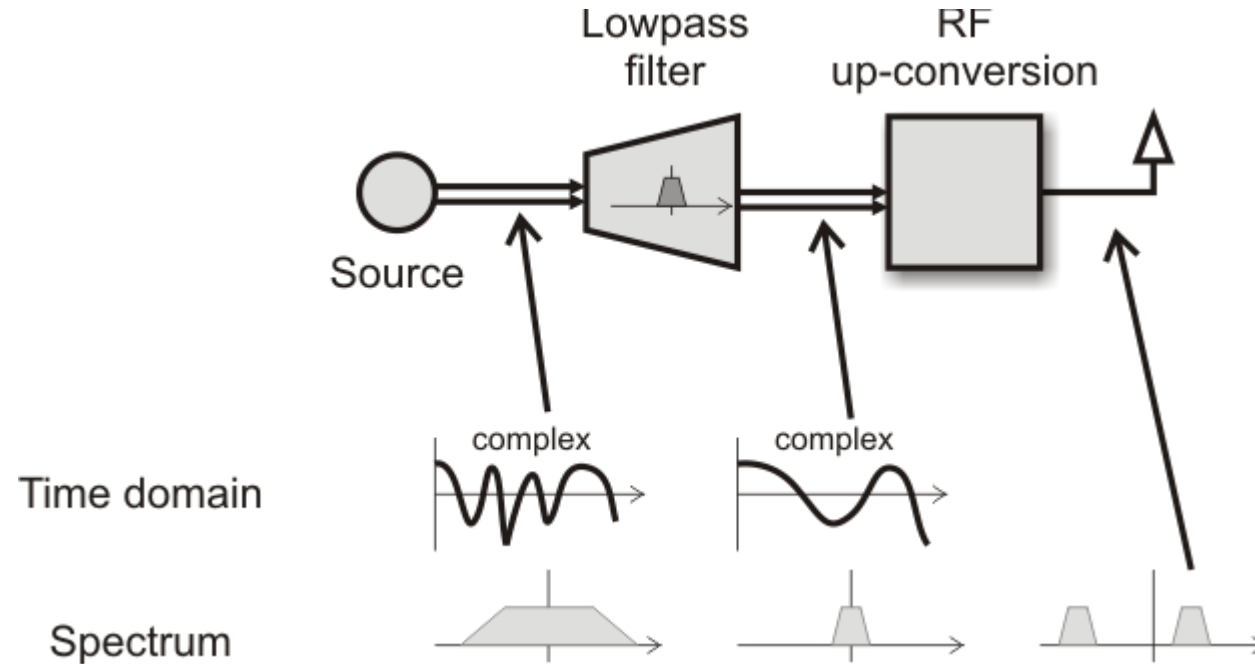
$$\begin{aligned}x(t) &= \Re\{\bar{x}(t)e^{j2\pi f_0 t}\} \\&= \Re\left\{\left(\bar{x}_I(t) + j\bar{x}_Q(t)\right)(\cos 2\pi f_0 t + j \sin 2\pi f_0 t)\right\} \\&= \bar{x}_I(t) \cos 2\pi f_0 t - \bar{x}_Q(t) \sin 2\pi f_0 t\end{aligned}$$

- $\bar{x}(t)$... Complex baseband signal (bandlimited)
 $\bar{x}_I(t), \bar{x}_Q(t)$... Real and imaginary parts of complex baseband signal
 f_0 ... Carrier frequency



Passband Modulation of Analog Signals

- A lowpass filter limits the bandwidth of the signal at the baseband
 - Filter defines also RF signal bandwidth



Digital modulation:

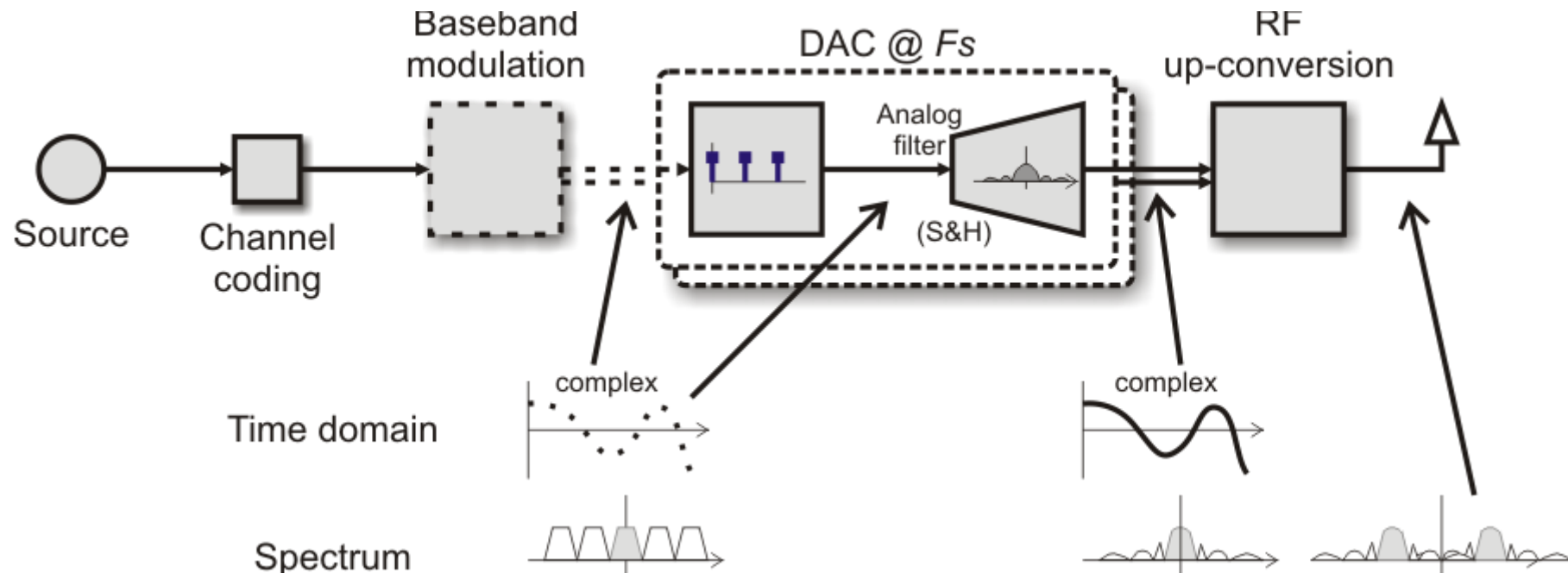
- Original signal is a sampled signal (complex symbols)
- Need to be translated into a continuous-time baseband signal prior to up-conversion

Baseband Modulation of Digital (Sampled) Signals

Baseband modulation at the symbol rate

- DAC performs conversion from discrete-time to continuous time
 - DAC = series of dirac pulses followed by a reconstruction filter (analog)
- Reconstruction filter = pulse shaping filter
 - **Little control over pulse shape (analog design)**
 - **Good image suppression: difficult and costly to realize**

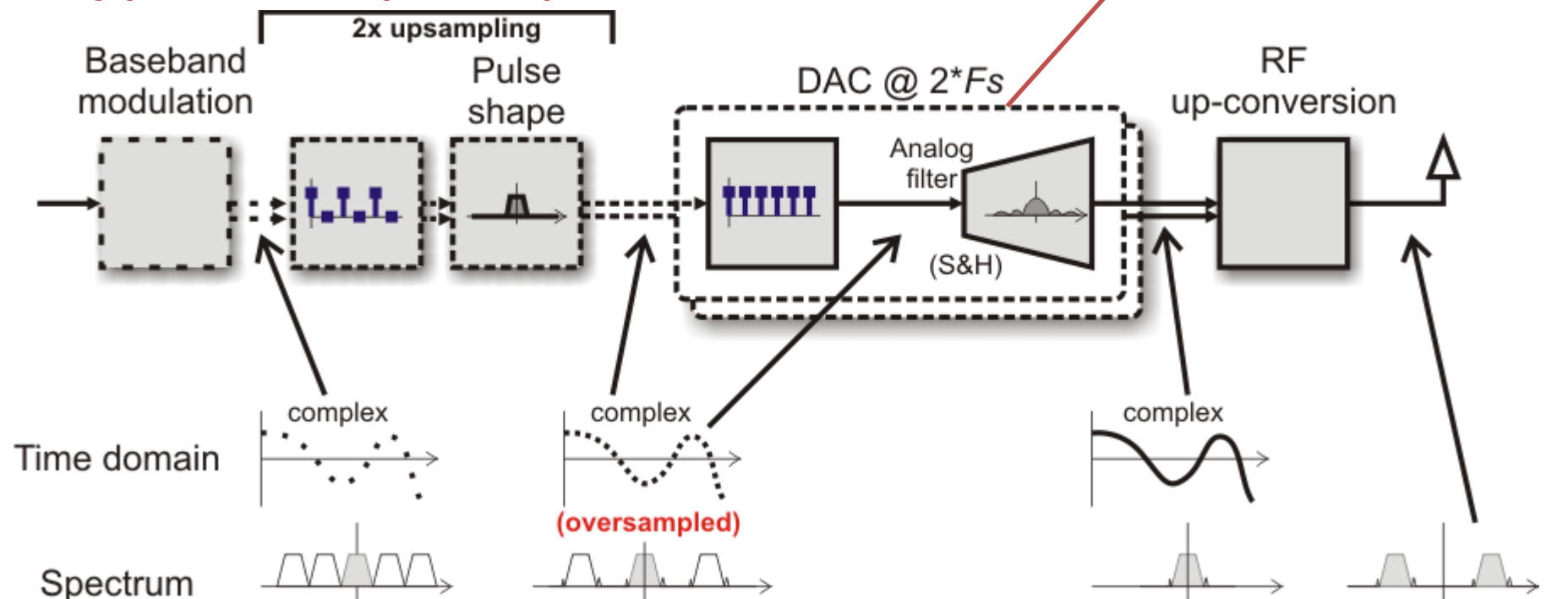
Need better control over image suppression and pulse-shape filter



The need for oversampling

Modulated baseband signal is upsampled in the digital domain

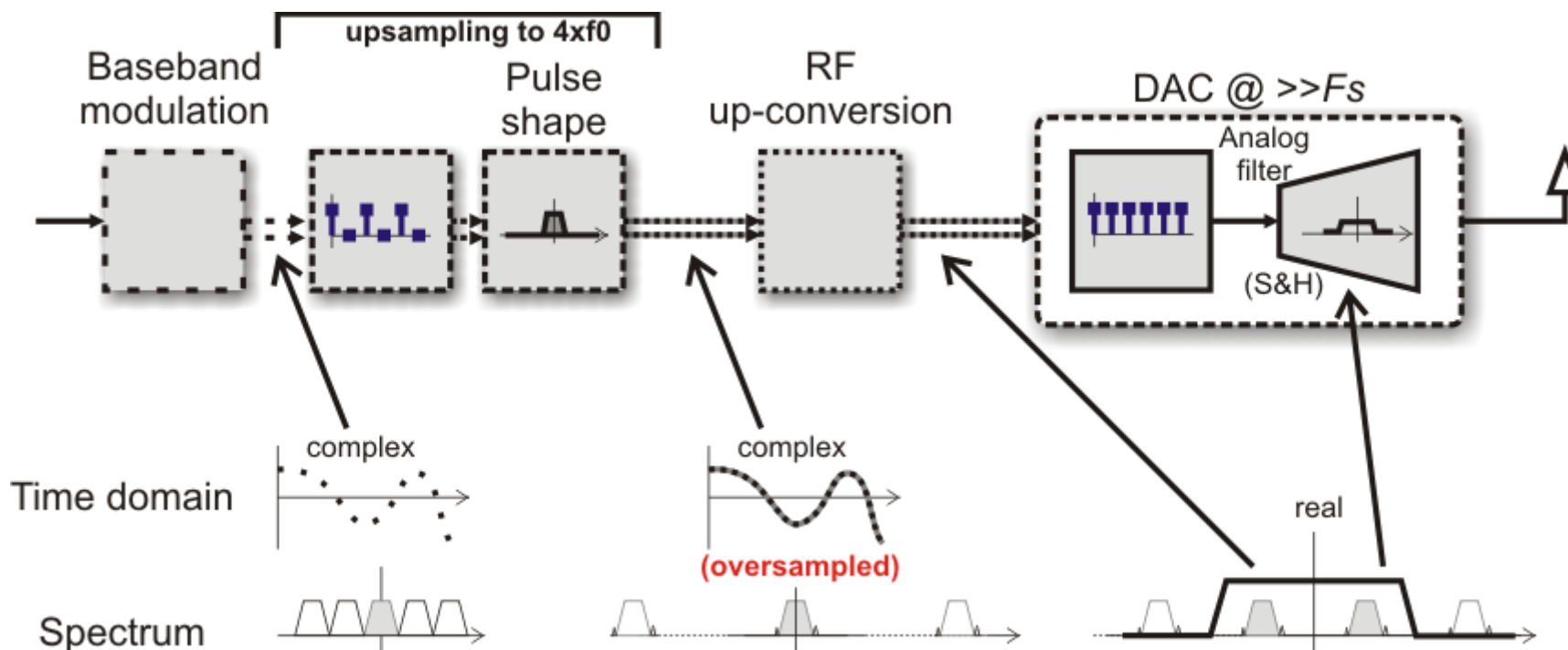
- Up-sampling ($\geq 2 \times$): zero-stuffing and filtering (image suppression)
- **Digital pulse shaping filter performs image suppression**
 - **Pulse shaping filter can be engineered carefully as needed**
- Images are pushed out to a higher frequency
 - **Relaxed analog (reconstruction) filter requirements**



All-Digital RF Transmitter (Software Radio)

Modulation to a carrier frequency in the digital (sampled) domain

- Signal is heavily up-sampled to $\geq 2(f_0 + BW_{BB})$ f_0 : carrier frequency
- Up-conversion is done in the digital domain
 - Note for sampling frequency of $4f_0$: $\sin 2\pi f_0 = \{0, +1, 0, -1\}$ and $\cos 2\pi f_0 = \{+1, 0, -1, 0\}$
- Images are easy to remove (very high frequency and at least $2f_0$ away)



Modulation for Passband Communication

PAM modulation: modulate the amplitude of a carrier with real-valued symbols a_k

$$s_{BP}(t) = \sum_k a_k g(t - kT) \cos(2\pi f_0 t)$$

PSK modulation: modulate the phase of a carrier with phasors ρ_k

$$s_{BP}(t) = \sum_k g(t - kT) \cos(2\pi f_0 t + \rho_k)$$

QAM modulation

- Modulate phase and amplitude of a carrier with pairs (a_k, ρ_k)

$$s_{BP}(t) = \sum_k a_k g(t - kT) \cos(2\pi f_0 t + \rho_k)$$

- Equivalently: modulate two orthogonal carriers with pairs $a_k = (\Re\{a_k\}, \Im\{a_k\})$

$$s_{BP}(t) = \sum_k \Re\{a_k\} g(t - kT) \cos(2\pi f_0 t) + \Im\{a_k\} \sin(2\pi f_0 t)$$

Baseband Signal Modulation for Passband Communication

PAM modulation: modulate the amplitude of a carrier with real-valued symbols a_k

$$s_{BP}(t) = \sum_k a_k g(t - kT)$$

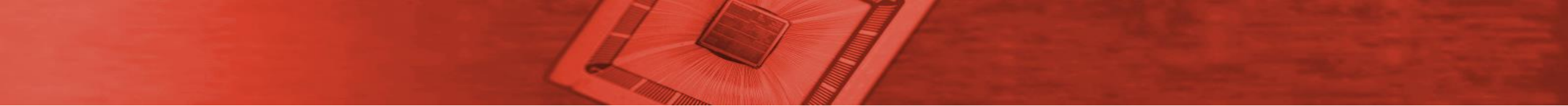
PSK modulation: modulate the phase of a carrier with phasors ρ_k

$$s_{BP}(t) = \sum_k e^{j\rho_k} g(t - kT)$$

QAM modulation

- Modulate two orthogonal carriers with pairs $a_k = (\Re\{a_k\}, \Im\{a_k\})$

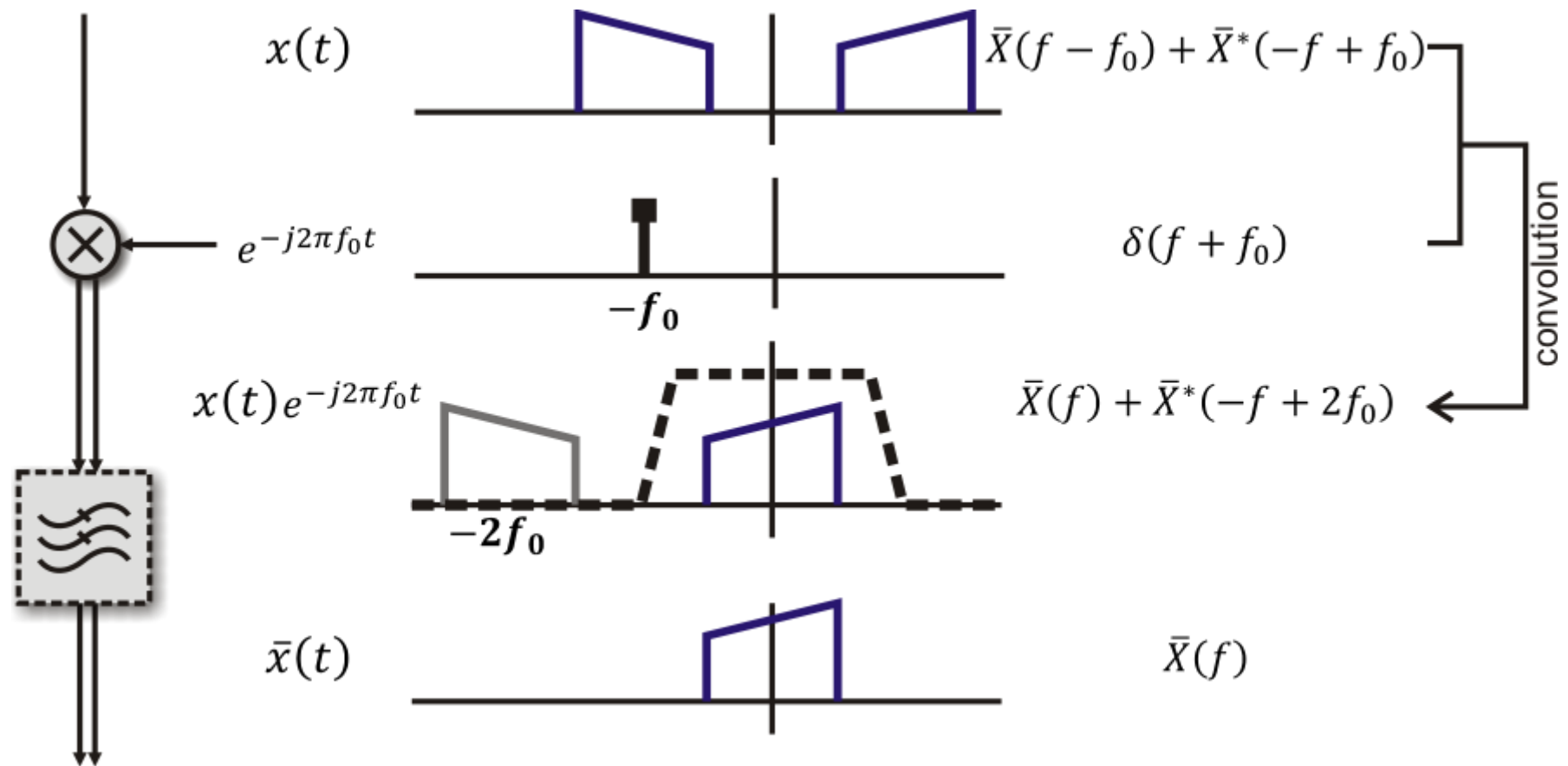
$$s_{BP}(t) = \sum_k (\Re\{a_k\} + j\Im\{a_k\}) g(t - kT)$$



Receiver

Direct Conversion Receiver

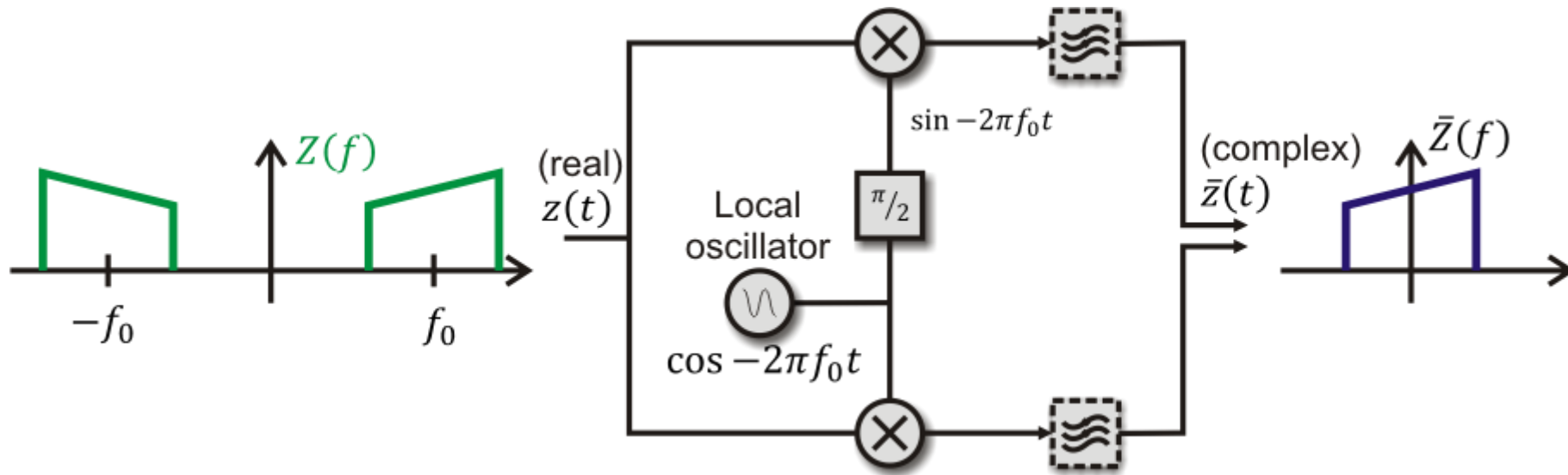
$$\bar{x}(t) = LP\{x(t)e^{-j2\pi f_0 t}\}$$



Down-Conversion

$$\begin{aligned}\bar{z}(t) &= LP\{z(t)e^{-j2\pi f_0 t}\} \\ &= LP\{z(t) \cos -2\pi f_0 t\} + jLP\{z(t) \sin -2\pi f_0 t\}\end{aligned}$$

- $z(t)$... Real-valued RF signal
 $\bar{z}(t)$... Complex baseband signal (bandlimited)
 f_0 ... Carrier frequency

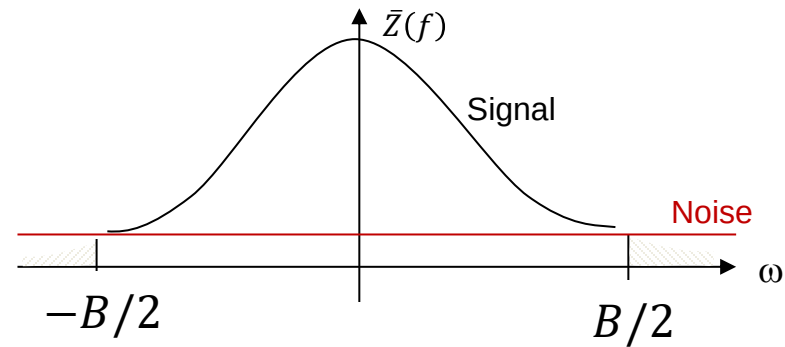


Bandlimited Approximation of Complex Baseband Signal

Consider a continuous time complex baseband signal $\bar{z}(t)$

- Approximate the baseband signal with a strictly bandlimited signal $\bar{z}_{BL}(t)$

$$\mathcal{E}\{|\bar{z}_{BL}(t) - \bar{z}(t)|^2\} = \int_{|f| > B/2} \bar{Z}(f) \rightarrow 0$$



- Useful transmitted passband signal had a bandwidth W

Baseband signal is limited to $[-B/2, B/2]$ with a lowpass filter

Bandlimited signal is a “sufficient statistics”

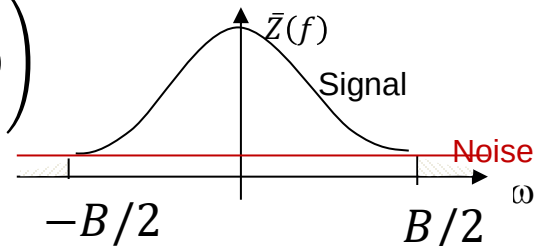
- By choosing B wide enough $W \leq B$, approximation to the useful signal $\mathcal{E}\{|\bar{z}_{BL}(t) - \bar{z}(t)|^2\}$ can be made arbitrarily small
- Only the noise is cut off. One can show that the noise outside signal bandwidth bears no information on the noise within the passband

Vectorspace Representation & Sampling

Vectorspace representation expresses a bandlimited continuous time signal as a superposition of delayed sinc pulses

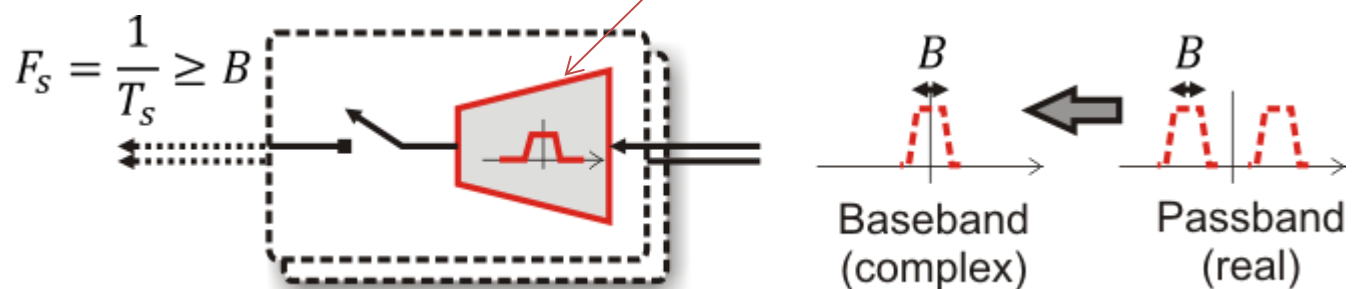
Base vector

Vectorspace components

$$\bar{z}_{BL}(t) = \sum_{n=-\infty}^{\infty} \bar{z}_{BL}[n] \operatorname{sinc}\left(\frac{\pi}{T_s}(t - nT_s)\right)$$


provided that $\frac{1}{T_s} \geq B$ with B: passband bandwidth

Vector space components are obtained by projecting the signal onto the sinc basis

$$\bar{z}_{BL}[n] = \int_{-\infty}^{\infty} \bar{z}_{BL}(\tau) \operatorname{sinc}\left(\frac{\pi}{T_s}(\tau - nT_s)\right) d\tau$$


$F_s = \frac{1}{T_s} \geq B$

Baseband (complex)

Passband (real)

Equivalence of Digital and Analog Signal properties

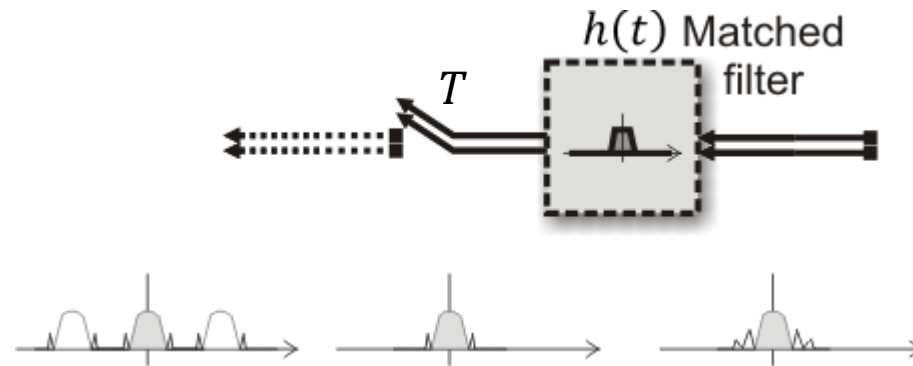
$$\begin{aligned} y(t) \big|_{t = kT_s} &= \int_{-\infty}^{\infty} h(kT_s - \nu) x(\nu) d\nu \\ &= T_s \sum_{n=-\infty}^{\infty} h(kT_s - nT_s) x(nT_s) \end{aligned}$$

Matched Filter

Consider a received signal with a pulseshape $g(t)$ **AWGN** with PSD N_0

$$\bar{z}(t) = \sum_k a_k g(t - kT) + n(t)$$

Matched filter $h(t)$: filter that maximizes the SNR
before sampling at the symbol rate T



How to choose $h(t)$ to maximize the SNR after sampling?

Matched Filter

Work in the frequency domain

- $H(f)$: matched filter frequency response
- $G(f)$: spectrum of the received signal (*here*: pulseshape of the transmitted signal)
- N_0 : power spectral density of the white noise

Noise power after the filter at time $t = T$

$$\sigma_n^2 = N_0 \int_{-\infty}^{\infty} |H(f)|^2 df$$

Signal power after sampling at $t = T$

$$P_s = \int_{-\infty}^{\infty} |H(f)G(f)|^2 df$$

Signal to noise ratio

$$\text{SNR} = \frac{\int_{-\infty}^{\infty} |H(f)G(f)|^2 df}{N_0 \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Matched Filter

Goal: maximize signal power P_s

Trick: use Cauchy-Schwarz inequality

$$\int_{-\infty}^{\infty} |H(f)G(f)|^2 df \leq \int_{-\infty}^{\infty} |H(f)|^2 df * \int_{-\infty}^{\infty} |G(f)|^2 df$$

- Left and right side are equal when $H(f) = \alpha G^*(f)$

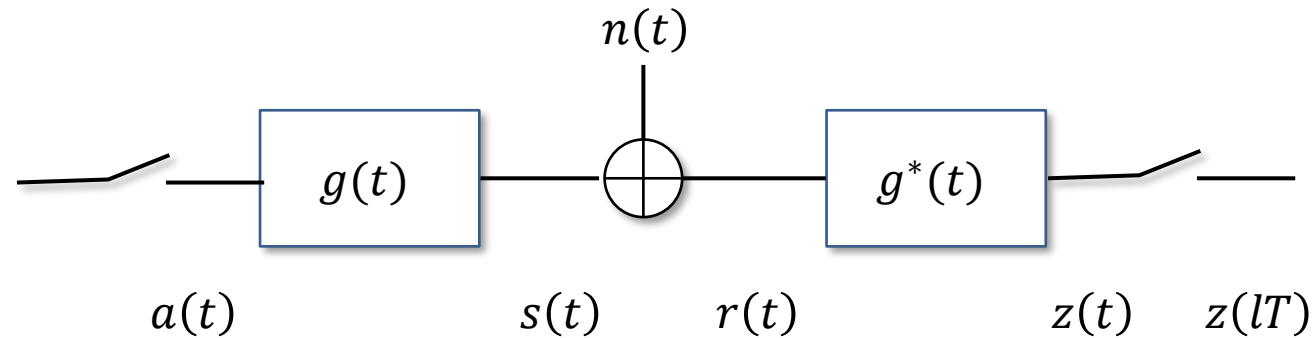
Therefore, to maximize P_s , we set $H(f) = \alpha G^*(f)$

Matched Filter

$$H(f) = \alpha G^*(f)$$

$$h(t) = g^*(-t)$$

Matched Filter Receiver (synchronized)



$$r(\tau) = \sum a_k g(t - kT)$$

$$z(t) = \int r(\tau) g^*(\tau - t) d\tau = \int \sum a_k g(\tau - kT) g^*(\tau - t) d\tau + \text{Noise}$$

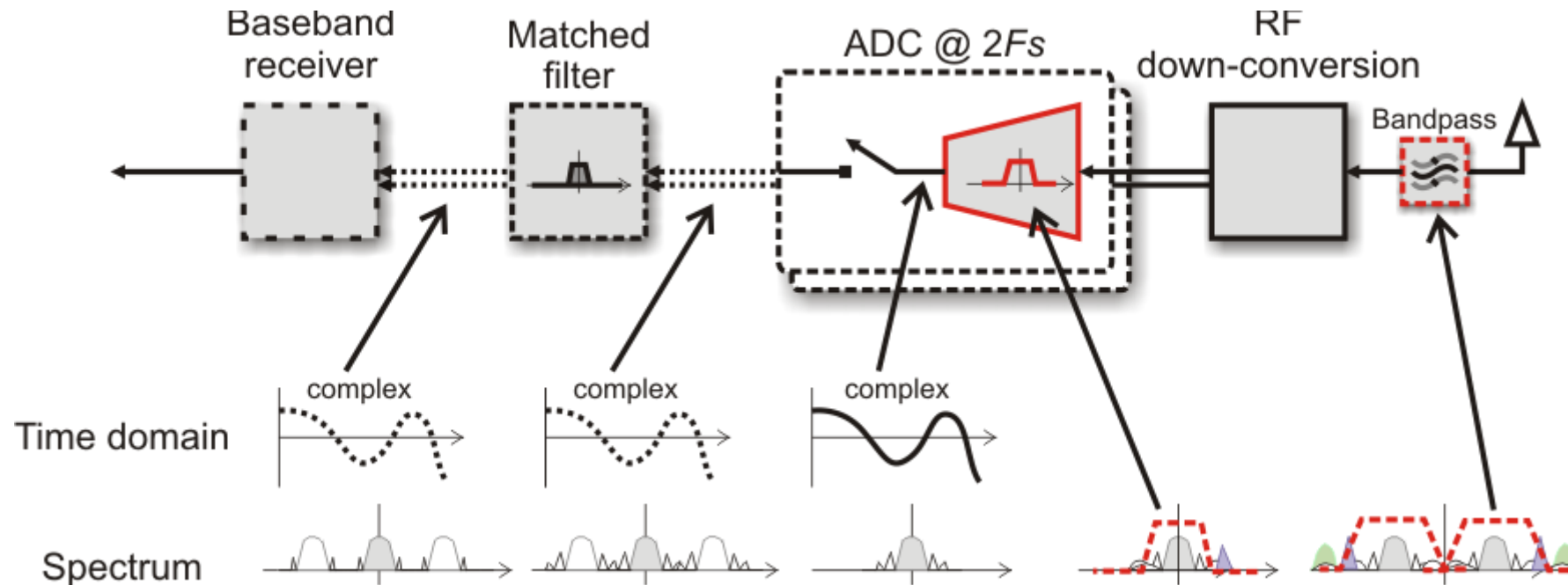
$$= \sum a_k \int g(\tau - kT) g^*(\tau - t) d\tau = \sum a_k g_{MF}(t - kT)$$

- With $g_F(t - kT) = \int g(\tau - kT) g^*(\tau - t) d\tau = \int g(\bar{\tau}) g^*(\bar{\tau} - (t - kT)) d\bar{\tau}$
- Transmit pulse shape and matched filter can be merged, but only for the analysis of the signal.

Direct Conversion Receiver Architecture

Down-conversion in the analog domain

- Received signal contains many (potentially strong) undesired signals
 - Bandpass filter isolates useful frequency range and removes blockers**
- ADC lowpass filter avoids aliasing of noise and remaining interferers
 - ADC often works with 2x oversampling**
 - Matched filter in the digital domain



Digital IF Receiver (Software Radio)

Down-conversion in the digital domain

- Received RF signal is low-pass filtered and sampled directly at $\text{RF} \geq 2(f_0 + BW_{BB})$
- Down-conversion on highly oversampled signal
 - Optional:** down-sampling to $\geq 2BW_{BB}$ before matched filter (reduce sampling rate) using LP-filter in down-conversion as anti-alias filter

